



discrete input module, Modicon X80,
16 inputs, 24V DC positive, for
severe environments

BMXDDI1602H

Product availability: Stock - Normally stocked in distribution facility

Main

Range of Product	Modicon X80
Product or Component Type	Discrete input module
Device Application	Industrial application in severe environment
Discrete input number	16
Discrete input voltage	24 V DCisolated
Discrete input logic	Positive logic (sink)

Complementary

Discrete input number	16 isolated conforming to IEC 60947-5-2 at 24 V DC 16 isolated conforming to IEC 61131-2 type 3 at 24 V DC
Input wiring mode	2 or 3 wires
Discrete input current	3.5 mA at 24 V DC
Voltage state 1 guaranteed	>= 11 V
Voltage state 0 guaranteed	< 5 V
Current state 1 guaranteed	>= 2 mA
Current state 0 guaranteed	<= 1.5 mA
Input impedance	6800 Ohm
Paralleling of inputs	Yes
Input protection type	Reverse polarity protection
Insulation resistance	> 10 MOhm 500 V DC
DC typical response time	4 ms
DC maximum response time	7 ms
Sensor power supply	19...30 V 60 °C 21.1...26.4 V 70 °C
Typical current consumption	76 mA DC
Current consumption	<= 107 mA 3.3 V DC
Power dissipation in W	2.5 W
Isolation between channels and bus	1500 V for 1 minute
Hot swapping fallback	Yes
MTBF reliability	775000 H
Voltage detection threshold	< 14 V DC sensor fault > 18 V DC sensor OK

Price is "List Price" and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Local signalling	1 LED (green) for module operating (RUN) 1 LED per channel (green) for channel diagnostic 1 LED (red) for module error (ERR) 1 LED (red) for module I/O
Height	4.08 in (103.7 mm)
Width	1.3 in (32 mm)
Depth	3.4 in (86 mm)
Product Weight	0.254 lb(US) (0.115 kg)

Environment

IP Degree of Protection	IP20
Protective treatment	Conformal coating
Marking	CE UKCA CULus RCM EAC
Product certifications	ATEX zone 2 IECEX zone 2 CCC Ex zone 2 (pending) CULus HazLoc
Standards	IEC 61131-2
Directives	2014/35/EU - low voltage directive 2014/30/EU - electromagnetic compatibility
Dielectric strength	1500 V AC 50/60 Hz 1 minute, primary/secondary
Ambient air temperature for operation	-13...158 °F (-25...70 °C)
Ambient air temperature for storage	-40...185 °F (-40...85 °C)
Relative humidity	5...95 % 131 °F (55 °C) without condensation
Operating altitude	0...6561.68 ft (0...2000 m) 2000...5000 m with derating
Pollution degree	2
Environmental characteristic	Gas resistant class Gx Gas resistant class 3C4 Dust resistant class 3S4 Sand resistant class 3S4 Salt resistant level 2 Mold growth resistant class 3B2 Fungal spore resistant class 3B2
Vibration resistance	3 gn
Shock resistance	30 gn

Ordering and shipping details

Category	US1PC3418160
Discount Schedule	PC34
GTIN	3595864025605
Returnability	Yes
Country of origin	US

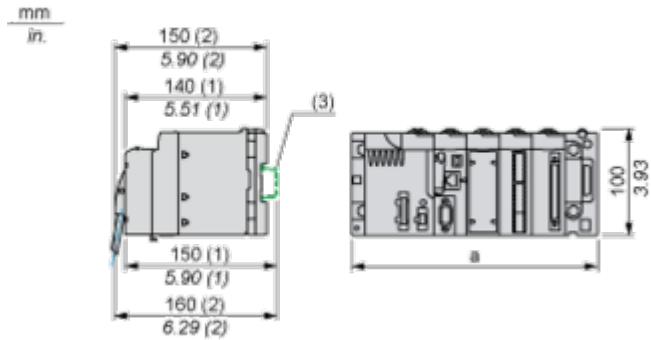
Packing Units

Unit Type of Package 1	PCE
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Dimensions Drawings

Modules Mounted on Racks

Dimensions



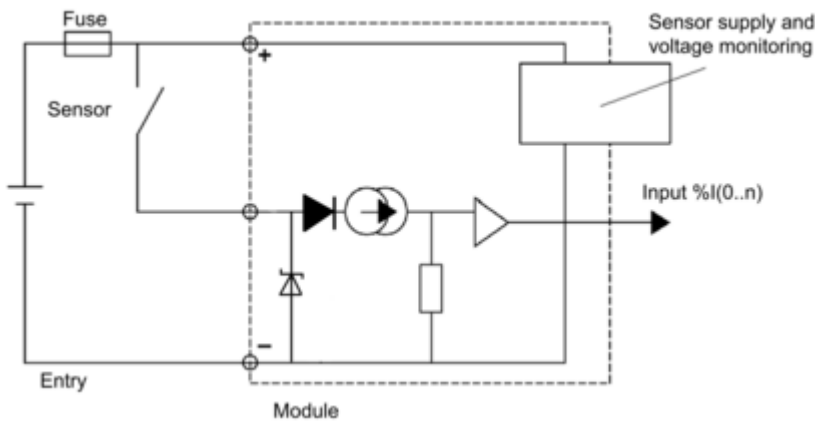
- (1) With removable terminal block (cage, screw or spring).
(2) With FCN connector.
(3) On AM1 ED rail: 35 mm wide, 15 mm deep. Only possible with BMXXBP0400/0400H/0600/0600H/0800/0800H rack.

Rack references	a in mm	a in in.
BMXXBP0400 and BMXXBP0400H	242.4	09.54
BMXXBP0600 and BMXXBP0600H	307.6	12.11
BMXXBP0800 and BMXXBP0800H	372.8	14.68
BMXXBP1200 and BMXXBP1200H	503.2	19.81

Connections and Schema

Connecting the Module

Input Circuit Diagram



Module Connection

